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Chapter 1

Introduction

The theory of Stochastic Calculus began as early as the 1880's with T.N.Thiele's work on time series and his model of Brownian motion and was followed by L.Bachelier's seminal work on Brownian motion in 1900 in an attempt to model the Paris Bourse. Bachelier's work is believed to have been the motivation for Itô discussed in the later sections. Soon after in 1905, A.Einstein's work to establish the molecular nature of matter modeled by Brownian motion and in 1827, R.Brown's observation of the random movement of pollen grains under a microscope provided empirical evidence of Brownian motion (See Figure 2.2). It's rather rare in Mathematics to have empirical evidence for a theory before it's proved mathematically and it was not until 1923, when N.Wiener mathematically proved the existence of Brownian Motion, that it was accepted by mathematicians.

The theory of Stochastic Differential Equations, in many respects, is similar to the theory of Ordinary Differential Equations as in the theorems or results which are often along the same lines as that of Ordinary Differential Equations, except that the complexity in this case is much more, as expected owing to the additional "noise" term. Moreover, the theory of Stochastic Differential Equations or Stochastic Calculus relies very heavily on the mathematically rich field of measure-theory and is inevitably used for many definitions and results. The similarities or dissimilarities in the approach between Stochastic and Ordinary Differential Equations and measure-theoretic results are mentioned throughout this paper wherever possible.

Thus, Stochastic Calculus in itself is a very mathematically rich domain, both from a point of view of pure and applied mathematics. Having started its journey with an application in Finance, its needless to say that it finds

prolific applications in many fields of study, Finance, Economics, Biology, Botany, Physics to name a few. One of its most notable application in recent times is the famous Black-Scholes formula obtained from the following stochastic differential equation

$$dS_t = S_t(\mu + \sigma dW_t) \quad (1.1)$$

where, $S_0 \geq 0$

Bachelier's earlier attempt of modeling the changes in the stock prices was similar but without the factor S_t in the above stochastic differential equation

Stochastic Differential Equation was also used heavily in Merton's paper on option pricing in 1976 and continues to be at the forefront of many researchers working on finance problems.

IN this paper, Chapter 2 focuses on laying the groundwork before plunging into more rigorous mathematical theorems in the later sections. As is known, by the Fundamental theorem of calculus, differentiation and integration are connected inseparably and hence Stochastic Integral is defined first. First, the most frequently used terms and their definitions, examples and important theorems are given from measure theory and analogues are drawn from Lebesgue-Stieltjes Integral in the construction of the Itô integral.

Chapter 3 provides some important theorems like the existence and uniqueness of solutions of Stochastic Differential Equations with examples, necessary and sufficient conditions for their existence. The two types of solutions, strong and weak solutions, are defined and illustrated by means of several famous examples. However, not all Stochastic Differential Equations have solutions. These SDEs are attempted to be solved by numerical methods which require great computation powers and is achieved by simulating many different sample paths of these equations to study the statistical properties of the solutions.

The various numerical methods, such as the Monte Carlo Simulation and the famous Euler Approximation methods are discussed in Chapter 4 and is illustrated with the help of a computer programs in MATLAB for large

number of simulations. The basic theory of these approximations is also discussed at some length.

Chapter 5 deals with the applications of SDEs in Economics and Biology respectively. In this chapter, firstly the basic groundwork laid by F.P.Ramsey in one of his most important papers, A Mathematical Theory of Saving, is discussed at some length to explore the transition of the problem tackled through an Ordinary Differential Equation by Ramsey to its Stochastic Counterpart, which gives rise to a special Stochastic Differential Delay Equation.

Finally, Chapter 6 is the concluding chapter and also mentions some interesting problems worth thinking where Stochastic Differential Equations could find usefulness for future consideration of these problems.

Chapter 2

Mathematical Preliminaries

Some of the most frequently used definitions are introduced here along with some important results and a few illustrations which will be used in the later sections. The definitions are broadly classified into three sections - basic measure theoretic definitions, Stochastic processes and the construction of the Itô integral. (See [2], [4], [8], [18], [22], [26])

2.1 Measure Theoretic Definitions

Definition 2.1.1 : σ - algebra

$\mathfrak{F}$, a collection of all subsets of a given set Ω is called a σ - algebra on a set Ω if the following conditions hold:

- (i) $\emptyset \in \mathfrak{F}$
- (ii) $\mathfrak{F}$ is closed under complementation i.e. $A \in \mathfrak{F} \implies A^C \in \mathfrak{F}$
- (iii) $\mathfrak{F}$ is closed under countable unions i.e. $A_i \in \mathfrak{F}, \forall i \implies \bigcup_{i=1}^{\infty} A_i \in \mathfrak{F}$

Using conditions (i)-(iii), its easy to see that a σ - algebra is also closed under countable intersections.

$\mathfrak{F}$ is called an **algebra** if conditions (i) and (ii) hold and $\mathfrak{F}$ is closed under finite unions i.e. $A_i \in \mathfrak{F}, \forall i = 1, \dots, n \implies \bigcup_{i=1}^n A_i \in \mathfrak{F}$. $(\Omega, \mathfrak{F})$ is called a measurable space and $(\Omega, \mathfrak{F}, \mu)$ is called a **measure space** where μ is a measure. If the measure μ is the probability measure, then $(\Omega, \mathfrak{F}, \mu)$ is called the probability measure space or probability space which we will encounter in the later sections.

Definition 2.1.2 : Filtration

An increasing sequence $\mathfrak{F}_n, \{\mathfrak{F}_n : n \in T\}$ of σ -algebras where $\mathfrak{F}_m \subset \mathfrak{F}_n$ for all $m \leq n, m, n \in T$, is called a filtration and $(\Omega, \mathfrak{F}_n, \mathfrak{F}, P)$ is a filtered probability space where P is the probability measure.

$\mathfrak{F}_n$ represents information known at time n and a filtration models the increase in our knowledge or information as time progresses but without any loss of information.

Definition 2.1.3 : Stochastic Process

A Stochastic Process is a collection of random variables $(X_t)_t$ on $(\Omega, \mathfrak{F}, P)$ and is called adapted if (X_t) is $\mathfrak{F}_t$ -measurable or $\mathfrak{F}_t$ -adapted i.e. X_t is known at time t .

Definition 2.1.4 : Martingales

A stochastic process $X = (X_t)$ is called a martingale with respect to the filtered probability space $(\Omega, \mathfrak{F}_n, \mathfrak{F}, P)$ if the following hold

- (i) X is adapted i.e. X_t is $\mathfrak{F}_t$ -measurable $\forall t \in T$
- (ii) $E[|X_t|] < \infty \forall t \in T$ i.e. $X_t \in L_1(P)$
- (iii) $E[X_s | \mathfrak{F}_t] = X_t$ P -a.s for $s \geq t, s, t \in T$

X is a sub-martingale if (i) and (ii) hold and $E[X_s | \mathfrak{F}_t] \geq X_t$ P -a.s for $s \geq t, s, t \in T$.

X is a super-martingale if (i) and (ii) hold and $E[X_s | \mathfrak{F}_t] \leq X_t$ P -a.s for $s \geq t, s, t \in T$.

Martingales model fair games, Sub-martingales model favourable games that is if the expectation at a later time given the information today is more than the realized value today and similarly Super-martingales model unfavourable games.

Definition 2.1.5 : Markov Process

A stochastic process $X = (X_t)$ is called a Markov Process with respect to the filtered probability space $(\Omega, \mathfrak{F}_n, \mathfrak{F}, P)$ if

- (i) X is adapted with respect to the filtration $\{\mathfrak{F}_n\}$
- (ii) X satisfies the Markov property i.e.

$$P(X_t | \mathfrak{F}_t) = P(X_t | X_s) \forall s, t \in T, s < t$$

Markov Processes can be thought to be memoryless processes in the sense that the future prediction depends only on the current state but not on the preceding states and can be used to model many natural phenomena. For example, credit ratings for companies released by agencies like Standard and Poor and Moody's assume that the probability of default of a company on its debts depends on whether the company defaulted the previous year and not on whether it defaulted 10 years before and hence is a Markov process and the ratings are calculated using the transition probabilities.

Theorem 2.1.6 : Every non-negative measurable function can be approximated by a sequence of simple functions i.e. for any measurable function $f \geq 0$, $\exists$ an increasing sequence of simple functions (ϕ_n) such that $(\phi_n) \rightarrow f$.

The above theorem is central to the definition of Lebesgue Integrals for defining the integral general functions. The construction of the Lebesgue Integral is done in stages; its defined first for the Indicator or the characteristic function, then the simple functions, which are linear combinations of indicator functions, followed by non-negative functions and finally for general functions.

Although Stochastic Integration is fundamentally different from Lebesgue Integration, the construction of the Stochastic Integral is defined analogously later in this chapter in Section 2.3.

2.2 Stochastic Processes and some Important results

Definition 2.2.1 : Poisson Process

A Stochastic Process N is called a Poisson Process if

(a) N is an adapted process with respect to $(\Omega, \mathfrak{F}_n, \mathfrak{F}, P)$, a filtered probability space

(b) N is a counting process i.e. $N = (N_t) : \Omega \rightarrow \mathbb{N} \cup \{\infty\}$ defined as $N_t(\omega) = \sum_{n \geq 1} \mathbb{1}_{\{t \geq T_n\}}$ where $\mathbb{1}_{\{t \geq T_n\}}$ is the indicator function and $(T_n)_{n \geq 1}$ is

a strictly increasing sequence of positive random variables.¹

- (c) (i) N has increments independent of the past i.e. $(N_t - N_s)$ is independent of $\mathfrak{F}_s$ for any s, t such that $0 \leq s < t < \infty$
- (ii) N has stationary increments i.e. the distribution of $(N_t - N_s)$ is same as that of $(N_v - N_u)$ for any s, t, u, v such that $t - s = v - u$, $0 \leq s < t < \infty$ and $0 \leq u < v < \infty$.

The Poisson process finds many applications, most notably in the insurance sector. For example, for a health insurance company, N_t represents the number of claims made up to time t and T_n would represent the arrival time of the claims from a person. Consequently, whether the n -th claim arrived or not has to be known to the company which is equivalent mathematically to T_n being a stopping time. However, the magnitude of the claims is not captured by modelling the above scenario by a Poisson process and to circumvent this limitation, the Compound Poisson process could be used to model the above.

A stochastic process $C = C_t$ is called a **Compound Poisson Process** if

$$C_t = Y_1 + Y_2 + \dots + Y_{N_t}$$

; $C_t = 0$ if $N_t = 0$

where N is a Poisson process and $(Y_i)_{i \geq 1}$ are independent and identically distributed (i.i.d) random variables independent of N .

Definition 2.2.2 : Brownian Motion

A Stochastic Process $B = (B_t)$ is called a one-dimensional Brownian Motion or the Wiener Process on the probability space $(\Omega, \mathfrak{F}, P)$ if

- (i) $B_0 = 0$ a.s and B is an adapted continuous process
- (ii) B has independent increments i.e. $(B_t - B_s)$ is independent of $\mathfrak{F}_s$ for any s, t such that $0 \leq s < t < \infty$
- (iii) B has Gaussian increments i.e. $(B_t - B_s)$ is normally distributed with mean 0 and variance $(t - s)$; $(B_t - B_s) \sim N(0, \sqrt{t - s})$.

$B = (B_t)_{0 \leq t < \infty}$ in $\mathbb{R}^n$ is called an N -dimensional Brownian motion if (i)

¹ N is adapted if and only if the corresponding random variables $(T_n)_{n \geq 1}$ are stopping times.

and (ii) hold and

(iii) for any s, t such that $0 \leq s < t$, $(B_t - B_s)$ is a multivariate normal random variable with mean 0 and co-variance matrix $(t - s)C$, where C is a given non-random matrix.

Among the many interesting properties of Brownian motion, the first is that Brownian motion is an everywhere continuous but nowhere differentiable function. Such examples are rare, the only other such functions being the Weierstrass function and the sum of any differentiable function and the Weierstrass function. Clearly from Figure 2.1(Left), it's easy to see that both the Wiener Process and the Weierstrass function are nowhere differentiable but everywhere continuous. Moreover, they're both fractals and the Weierstrass function is, in fact, a self similar process i.e. if a sub part of the graph is zoomed in, it exactly resembles the original graph as shown in Figure 2.1 (Right). The corresponding MATLAB codes can be found in Appendix M1.

Figure 2.1: Time evolution of One-Dimensional Brownian Motion (Left) and the Weierstrass function(Right)

Brownian motion was first observed by Dr Robert Brown, while looking at some pollen grains suspended in water under a microscope and found the pollen grains to be in a constant jittery motion, as represented in Figure 2.2 (See Appendix M2 for the code), and hence called Brownian motion which was established mathematically much later by N. Wiener.

Figure 2.2 : Two-Dimensional Brownian Motion

Definition 2.2.3 : Lévy process

A Stochastic process $X = (X_t)$ is called a Lévy process if

- (i) X is an adapted process such that $X_0 = 0$ a.s
- (ii) X has increments independent of the past i.e. $(X_t - X_s)$ is independent of $\mathfrak{F}_s$ for any s, t such that $0 \leq s < t < \infty$
- (iii) X has stationary increments i.e. $(X_t - X_s)$ has the same distribution as X_{t-s} for any s, t such that $0 \leq s < t < \infty$
- (iv) X_t is continuous in probability i.e. $\lim_{t \rightarrow s} X_t = X_s$, where the limit is taken in probability.

The Poisson Process and Brownian motion, both being particular cases of Lévy process, are the two most important continuous time stochastic processes.

Definition 2.2.4 : Local Martingale

A Stochastic process $X = (X_t)$ is called a local martingale if for some localizing sequence of stopping times $\{\tau_n\} \rightarrow \infty$, $X_{(t \wedge \tau_n)} = X_{\min(t, \tau_n)}$ is a martingale.

Definition 2.2.5 : Semi Martingale

A Stochastic process $X = (X_t)$ is called a Semi Martingale if X_t can be decomposed as the sum of a Local Martingale and an adapted process which is locally of finite variation i.e. $X_t = M_t + A_t$, where M_t is a Local Martingale and A_t is an adapted locally of finite variation process.

Theorem 2.2.5 : Doob's Decomposition Theorem

Let $X = (X_t)$ be an adapted Stochastic process such that $X_n \in L^\infty$. Then X can be decomposed as the sum of a martingale null at zero and a predictable process null at zero i.e.

$X_t = M_t + A_t$; where M_t is a martingale null at zero and A_t is a predictable process null at zero.

Remark : One of the important theorems in measure theory is the Lebesgue's Decomposition theorem which states the following :

Given two σ -finite signed measures μ and ν , $\exists$ two signed measures ν_0 , absolutely continuous to μ , and ν_1 , singular to μ , such that $\nu = \nu_0 + \nu_1$.

Lévy-Itô's Decomposition Theorem is the analogue of this theorem in Stochastic Calculus but it will not be used in this paper and is hence not given here. (See [4] and [22] for further reading)

2.3 Stochastic Integration and Differentiation

Stochastic Integration was introduced by Kiyosi Itô in 1944 and although it is fundamentally different from Riemann-Stieltjes Integration, its helpful to have a background in Riemann-Stieltjes Integration since the construction of the stochastic integral is similar to the Riemann-Stieltjes and Lebesgue

Integral, as discussed after Theorem 2.1.6.

Unlike Riemann or Lebesgue-Stieltjes Integration, where the integrand was a continuous function, or discontinuous with the set of points of discontinuities having measure zero, and integrator being the difference of two increasing functions which are locally of finite variation, both the integrator and integrand in Stochastic Integration are stochastic processes.

For example, the Riemann Integration with both the integrator and the integrand being x gives us the following

$$\int_0^t x \, dx = \frac{t^2}{2}$$

. Contrast this with the Stochastic integration where both the integrator and the integrand being the Wiener Process/ Brownian motion gives us,

$$\int_0^t B_s dB_s = \frac{(B_t)^2}{2} - \frac{t}{2}$$

. The deterministic part $\frac{(B_t)^2}{2}$ is what one would expect to obtain by ordinary Leibniz/ Newton calculus but due to the noise term, we have an additional term $\frac{t}{2}$ which is the Quadratic Variation term given by definition 2.3.4.

Definition 2.3.1 : Simple Function

A stochastic process X is called simple if $\exists$ a partition $0 = t_0 < t_1 < t_2 < \dots < t_n = T < \infty$ and X_i be an uniformly bounded random variable where $\{X_i\}$ is $\mathfrak{F}_{t_i}$ - measurable such that $|X_{t_i}(w)| \leq c \forall w, i = 0, 1, 2, 3, \dots, n$ for some constant, c and if $X(t, w)$ can be expressed as

$$X(t, w) = X_{\{0\}}(w)I_0(t) + \sum_{i=0}^n X_i(w)I_{(t_i, t_{i+1}]}(t), 0 \leq t \leq T, w \in \omega$$

Definition 2.3.2 : Martingale Transform

If $X = (X_n)$ is a martingale and $C = (C_n)$ is a predictable process, then the Martingale Transform of X by C is given by

$$(C.X)_n = \sum_{k=1}^n C_k[X_k - X_{k-1}]$$

Definition 2.3.3 : Itô Process

A stochastic process $X = (X_t)$ is called an Itô process or stochastic integral

if its of the form

$$X_t(w) = X_0(w) + \int_0^t u(s, w)ds + \int_0^t v(s, w)dB_t \quad (2.1)$$

where $B = (B_t)$ is Brownian motion

Definition 2.3.4 : Quadratic Covariation

For stochastic processes X and Y satisfying (2.1), $< X, Y >$ is called the Quadratic Covariation and is given by

$$< X, Y >_t = X_t Y_t - X_0 Y_0 - \int_0^t X(s, w) dY(s, w) - \int_0^t Y(s, w) dX(s, w)$$

For a stochastic process X , if $\lim_{t \rightarrow \infty} \Pi_n X$ exists and is finite over $[0, t]$, then X is said to have finite quadratic variation over $[0, t]$ denoted by

$$< X >_t = \lim_{t \rightarrow \infty} \Pi_n X = \sup_{\Pi_n} \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i})^2$$

where Π_n is a partition of $[0, t]$ and is given by $\Pi_n X = \sum_{t_i \in \Pi_n} (X_{t_{i+1}} - X_{t_i})^2$. The Quadratic variation is analogous to finite variation for Riemann-Lebesgue Integration.

Construction of the Itô Integral

When both the Integrand and the integrator are stochastic processes, its beyond the reach of ordinary Leibniz/Newton Calculus to define integration or differentiation of such processes and is done by Itô integration or stochastic integration.

Itô Calculus gives meaning to integrals of the form

$$\int_0^t X dY \text{ or } \int_0^t X(s, w) dY(s, w), s \in T, w \in \omega$$

, where both X and Y are stochastic processes. If the integrator, Y is a finite variation increasing process and X is a continuous jointly measurable process, then the integral, $\int_0^t X(s, w) dY(s, w)$ can be represented as the limit of sums i.e.

$$\int_0^t X(s, w) dY(s, w) = \lim_{n \rightarrow \infty} \sum_{t_k, t_{k+1} \in \Pi_n} X_{s_k} (Y_{t_{k+1}} - Y_{t_k}) a.s$$

for $t_k \leq s_k \leq t_{k+1}$ where (Π_n) is a sequence of finite random partitions of $[0, t]$ such that $\text{mesh}(\Pi_n) \rightarrow 0$ as $n \rightarrow \infty$.

If X and Y are both continuous processes, then we could define the integral using approximating Riemann sums but if Y is of infinite variation, the Riemann sum will not exist. Hence defining the integral using Riemann sums is feasible only if Y is of finite variation.

Stochastic Integration addresses this shortcoming and we can thus define Itô Integration for processes or integrators of infinite variation. The most frequently used integrator is the Wiener Process or Brownian motion which has paths of infinite variation. While defining the Itô integral, the set of integrands have to be restricted to adapted processes, whereas no such restriction is required when defining the Lebesgue-Stieltjes integral with respect to processes of finite variation.

The Integral is first defined for the integrator being the Wiener process and the integrator being the indicator functions and then extended to other functions similar to the approach followed for construction of the Lebesgue-Stieltjes Integral.

Case - I When the integrand, X is an indicator function, $X = I_{(a,b]}(t)$

$$\int_0^t X dY = \int_0^t X(s, w) dB(s, w) = \begin{cases} 0, & \text{if } t < a. \\ B(t) - B(a), & \text{if } a \leq t \leq b. \\ B(b) - B(a), & \text{if } b \leq t. \end{cases}$$

Case - II When the integrand, X is a simple function

From Definition 2.3.1, X can be represented as

$$X(t, w) = X_{\{0\}}(w)I_0(t) + \sum_{i=0}^n X_i(w)I_{(t_i, t_{i+1}]}(t), 0 \leq t \leq T, w \in \omega$$

,

Hence the Itô integral is

$$I_t(X) = \int_0^t X dB = \sum_{i=0}^{k-1} X_i(w)[B(t_{i+1}, w) - B(t_i, w)] + X_k(w)[B(t, w) - B(t_k, w)]. \quad (2.2)$$

It's easy to observe that the Itô integrals for simple integrands is actually the martingale transform of B by X and that the Itô Integral is linear and a continuous martingale.

Before, we can define the Itô integral for more general integrands, we need the following properties :

Lemma 2.3.1 : Itô Isometry

$$E[(\int_0^t X(s, w) dW(s, w))^2] = E[\int_0^t X(s)^2 ds]$$

The Itô Isometry translates to $(dW(t))^2 = dt$, surprising as it may appear. The proof of the Itô Isometry is given at the beginning of the next chapter.

Case - III When the Integrand X is such that $\int_0^t E[(X(u))^2] du < \infty$

Just as in defining the Riemann or the Lebesgue-Stieltjes Integral, we want a sequence of functions or stochastic processes $X_n(t, w)$, in this case, approximating X , $X_n(t, w) \rightarrow X(t, w)$.

The Itô Isometry implies that integration will be defined for a process X if

$$\int_0^t E[(X(u))^2] du < \infty \quad (2.3)$$

If (2.3) does not hold, then the Itô integral, $I_t(X)$ yields a local martingale, which are more general than true martingales, but not a martingale.

We now need a class of integrands which can be approximated by a sequence of simple integrands. This class is $(\beta([0, \infty)), X, \mathfrak{F})$ - measurable $\mathfrak{F}_t$ - adapted processes X such that (2.1) holds for all t . Given any adapted predictable process X , $\exists$ a sequence of simple predictable processes X_n such that

$$\int_0^t E[\{X(s, w) - X_n(s, w)\}^2] ds \rightarrow 0$$

as $n \rightarrow \infty$. So $I_t(X_n)$ is a square integrable random variable for all n .

By Itô's Isometry, we have

$$E[\int_0^t \{(X(s, w) - X_n(s, w)) dB(s, w)\}^2] \rightarrow 0$$

as $n \rightarrow \infty$ or

$$E\left[\int_0^t \{I_t(X - X_n)\}^2\right] \rightarrow 0$$

as $n \rightarrow \infty$.

Thus $(I_t(X_n))$ is a Cauchy sequence in L^2 and we know L^2 is complete, hence $(I_t(X_n)) \rightarrow (I_t(X)) \in L^2$. We also need the Itô integral to be continuous so that we can use some form of dominated convergence theorems.

$$\int_0^t X(s, w) dB(s, w) = I_t(X) = \lim_{n \rightarrow \infty} \int_0^t X_n(s, w) dB(s, w)$$

Also, in the case of general integrands too, the Itô integral is linear. The above construction of the Itô integral was for the Wiener process as the integrator, which is what is used most frequently, but the most general integrator is a semi-martingale.

Much like Riemann or Lebesgue-Stieltjes Integration, while solving the integration problems, we hardly ever use the theoretical definitions of Integration and stochastic integration is no different. Most of the SDEs are solved using various other techniques, in fact, not all the SDEs are solvable explicitly and are often solved numerically.

Chapter 3

Theory of SDEs

An ordinary Differential Equation (ODE) is represented as follows

$$\frac{dX}{dt} = u(t)$$

The general form of a Stochastic Differential Equation (SDE) is

$$dX_t = u(t, X_t)dt + v(t, X_t)dW_t$$

, where W_t is the Wiener Process or Brownian motion, $u(t, X_t)$ is called the drift function or the deterministic part and $v(t, X_t)$ is called the diffusion function or the random part. Its much harder to solve SDEs than ODEs and its often the case that they are not even solvable explicitly theoretically and have to be solved numerically which is discussed in chapter 4. (The definitions and theorems in this chapter are due to [2], [4], [8], [13], [15], [18], [22] and [26] and we denote Brownian motion or the Wiener process by $W = W_t$ hereafter)

3.1 Itô's Formula

We first prove the Itô Isometry, the statement of which was given in section 3 of chapter 1 as this forms the crux of the following section.

Proof of Lemma 2.3.1

We have to prove $E[(\int_0^t X(s, w)dW(s, w))^2] = E[\int_0^t X(s)^2 ds]$.
From equation 2.2, we have

$$E[(\int_0^t X(s, w)dW(s, w))^2]$$

$$= E[\{\sum_{i=0}^{k-1} X_i(w)[W(t_{i+1}, w) - W(t_i, w)] + X_k(w)[W(t, w) - W(t_k, w)]\}^2]$$

Squaring the sum above gives us two square terms and the cross terms and since expectation is linear,

$$= E[\{\sum_{i=0}^{k-1} X_i^2(w)[W(t_{i+1}, w) - W(t_i, w)]^2 + X_k^2(w)[W(t, w) - W(t_k, w)]^2\}]$$

since $E[\{W(t_{i+1}, w) - W(t_i, w)\}\{W(t, w) - W(t_k, w)\}] = 0$ as W is the Wiener process. Thus we have,

$$\sum_{i=0}^{k-1} E[X_i^2(w)]E[\{W(t_{i+1}, w) - W(t_i, w)\}^2] + E[X_k^2(w)]E[\{W(t, w) - W(t_k, w)\}^2]$$

as X_i is $\mathfrak{F}_{t_i}$ - measurable and X_i^2 is independent of $\{W(t_{i+1}, w) - W(t_i, w)\}^2$. Also, $\{W(t_{i+1}, w) - W(t_i, w)\}$ has variance $t_{i+1} - t_i$. Hence,

$$= \sum_{i=0}^{k-1} \{E[X_i^2(w)](t_{i+1} - t_i) + E[X_k^2(w)](t - t_k)\}$$

Since expectation is an integral itself over Ω , using Fubini's theorem (for interchanging the integrals), we get

$$\int_0^t E[X(s)^2]ds = E\{\int_0^t X(s)^2 ds\}$$

$$\int_0^t E[X(s)^2]ds = \sum_{i=0}^{k-1} \{E[X_i^2(w)](t_{i+1} - t_i) + E[X_k^2(w)](t - t_k)\}$$

Hence Proved.

Theorem 3.1.1 : Basic Itô Formula

If $X = (X_t)$ is a Stochastic process such that $dX_t = u(t)dt + v(t)dW_t$, $X_0 = x_0$ and $f \in C^2$ i.e. f is a real twice-differentiable function and the derivative is also continuous, then

$$f(X_t) = f(x_0) + \int_0^t f'(X_s)dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s$$

, where $\langle X \rangle$ is the Quadratic Variation of X
or in differential form as,

$$df(X_t) = f'(X_t)dX_t + \frac{1}{2}f''(X_t)d\langle X \rangle_t$$

.

Theorem 3.1.2 : Itô's Formula

If $X = (X_t)$ is a Stochastic process such that $dX_t = u(t)dt + v(t)dW_t$, $X_0 = x_0$
and $f = f(t, X_t) \in C^{1,2}$ i.e. f is continuously differentiable once in t and twice
in X_t , then

$$f(t, X_t) = f(0, x_0) + \int_0^t (f_s(s, X_s) + u(s)f_X(s, X_s) + \frac{1}{2}v(s)^2 f_{XX}(s, X_s))ds + \int_0^t v(s)f_X(s, X_s)dW_s \quad (3.1)$$

or in differential form as,

$$df(t, X_t) = \{f_t(t, X_t) + u(t)f_X(t, X_t) + \frac{1}{2}v(t)^2 f_{XX}(t, X_t)\}dt + v(t)f_X(t, X_t)dW_t$$

.

Let us see an example of a frequently used Ordinary Differential equation,
the Malthusian growth model, and its stochastic counterpart.

$$\frac{dX(t)}{X(t)} = \alpha dt, X(0) = X_0$$

where $X(t)$ denotes the population at time t , α is the relative rate of growth
and X_0 is the initial population. Solving the above ODE is rather straight-
forward and the solution is given by $X(t) = X_0 \exp \alpha t$.

The parameter α may not be completely known and there could be vari-
ous unexpected factors which could influence the relative rate of growth and
to take the randomness arising due to unknown factors into consideration,
we would have a deterministic part and a "noise" term in α which models
the randomness.

Example 3.1.1

Consider the Stochastic counterpart of the above ODE given by

$$\frac{dX(t)}{X(t)} = b(t)dt, b(t) = C_t + \alpha W_t, X(0) = X_0$$

where $C_t = c$ and α are constants, W_t is the Wiener Process and X_0 is the initial population and $b(t)$ models the randomness.

Itô's Interpretation leads to the following SDE

$$\frac{dX(t)}{X(t)} = cdt + \alpha dW_t$$

$$dX_t = (cX_t)dt + (\alpha X_t)dW_t$$

The solution is $X_t = X_0 \exp(c - \frac{\alpha^2}{2})t + \alpha W_t$ obtained exactly as done below in example 3.1.2.

Example 3.1.2

Consider the Stochastic Differential Equation,

$$dS_t = (\mu S_t)dt + (\sigma S_t)dW_t,$$

S_0 is the initial value and represents the initial stock price and $S = (S_t)$ denotes the log return of prices at time t , μ denotes the mean and σ denotes the volatility.

Let $f(t, S_t) = \log(S_t) \in C^{1,2}, S_t > 0$. We have $u(t, S_t) = \mu S_t$ and $v(t, S_t) = \sigma S_t$.

By theorem 3.1.2,

$$f(t, S_t) = f(0, S_0) + \int_0^t (f_s(s, S_s) + \mu S_s f_S(s, S_s) + \frac{1}{2}(\sigma S_s)^2 f_{SS}(s, S_s))ds + \int_0^t \sigma S_s f_S(s, S_s)dW_s$$

Substituting $f(0, S_0) = \log(S_0)$, $f_s(s, S_s) = 0$, $f_S(s, S_s) = \frac{1}{S_s}$, $f_{SS}(s, S_s) = -\frac{1}{S_s^2}$ in above equation, we get

$$\log(S_t) = \log(S_0) + \int_0^t \left\{ \mu - \frac{\sigma^2}{2} \right\} dt + \int_0^t \sigma dW_s$$

$$S_t = S_0 \exp \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t$$

is the required solution of SDE and is called a Geometric Brownian motion.

3.2 Existence and Uniqueness Theorem

Once we know how to solve a SDE,

$$dX_t = u(t, X_t)dt + v(t, X_t)dW_t, X_0 = x_0,$$

the next obvious questions are whether we can have Existence and Uniqueness theorems on similar lines to the theory of Ordinary Differential Equations. It turns out, we do have existence and uniqueness theorems but as expected, its much more complicated than the theory for ODEs owing to the fact that we now have random processes.

The Picard's iterative scheme for the above SDE is given by

$$X_t^0 = x_0, X_t^{n+1} = x_0 + \int_0^t u(s, X_s^n)ds + \int_0^t v(s, X_s^n)dW_s$$

and is used to prove theorem 3.2.1. An immediate observation is the similarity of the Picard's scheme to that defined for ODEs except the random term. If a solution to the given SDE exists and if the above sequence converges to say X , then X could be a solution.

Theorem 3.2.1 : Existence and Uniqueness theorem for SDEs

A stochastic Differential Equation $dX_t = u(t, X_t)dt + v(t, X_t)dW_t$ has a unique solution $X_t(w)$ such that

- (i) X is $\mathfrak{F}_t^W$ - adapted where $\mathfrak{F}_t^W$ is the natural filtration of the Wiener Process, W
- (ii) X is a continuous Markov Process
- (iii) $\exists$ a constant C such that $E[X_t^2] \leq C \exp \{Ct\}(1 + x_0^2)$

if the following conditions hold

- (i) the local Lipschitz condition i.e.

$$|u(t, x) - u(t, y)| + |v(t, x) - v(t, y)| \leq D|x - y|$$

where D is a constant and $x, y \in \mathbb{R}^n, t \in T$

- (ii) the linear growth condition i.e.

$$|u(t, x)| + |v(t, x)| \leq D(1 + |x|)$$

for $x, y \in \mathbb{R}^n, t \in T$.

There are two types of solutions to an SDE, a strong solution and a weak solution.

Definition 3.2.1 : Strong Solution

A Stochastic process $X = (X_t)$ is called a strong solution of (3.1) if X_t is $\mathfrak{F}_t^W$ - adapted, which is the natural filtration of the Wiener Process.

Its obtained by solving the differential equation for each sample path of the stochastic process as a function i.e. solving the SDE path wise since the SDE is defined in this case within a filtered probability space and is solved there. Clearly, a solution X given by Theorem 3.2.1 will be a strong solution.

Its often the case that a SDE has no strong solution, probably because of the restrictive conditions of the drift and diffusion coefficients, and hence we need another form of a solution.

Definition 3.2.2 : Weak Solution

A stochastic process X is called a weak solution of (3.1) if we work only with the drift and diffusion coefficients and need a pair of processes $((X_t, W_t), \mathfrak{F}_t)$ on a probability space $(\Omega, \mathfrak{F}, P)$.

Its obtained by setting up a filtered probability space and defining and solving the SDE on that probability space and happens to be more appropriate in many situations.

Example 3.2.1

Consider the SDE given by

$$dX_t = \left\{ \frac{-X_t}{(1-t)} \right\} dt + dW_t, X_0 = 0$$

over the interval $[0, T], T < 1$.

We have $u(t, X_t) = \frac{-X_t}{(1-t)}$ and $v(t, X_t) = 1$. Clearly, u and v satisfy the Lipschitz and linear growth conditions and hence by theorem 3.2.1, there must exist an unique solutions for the given SDE.

Let $f(t, X_t) = \frac{X_t}{(1-t)}$.

$$f_t = \frac{X_t}{(1-t)^2}, f_X = \frac{1}{(1-t)}, f_{XX} = 0$$

By theorem 3.1.2,

$$\frac{X_t}{(1-t)} = \int_0^t \left\{ \frac{X_s}{(1-s)^2} + \frac{-X_s}{(1-s)} \frac{1}{(1-s)} \right\} ds + \int_0^t \frac{1}{(1-t)} dW_s$$

$$X_t = (1-t) \int_0^t \frac{1}{(1-s)} dW_s$$

is the solution of the SDE above, also called the Brownian Bridge.

Example 3.2.2

Consider the SDE

$$dX_t = (aX_t + b_t)dt + \sigma_t dW_t, X_0 = x_0$$

, a is a constant, $|b_t| \leq M < \infty, |\sigma_t| \leq N < \infty$ such that the integrals are well-defined.

We have $u(t, X_t) = (aX_t + b_t), v(t, X_t) = \sigma_t$. Taking $D = M + N + |a|$ in theorem 3.2.1, the locally Lipschitz condition and the linear growth condition hold and hence there must be a unique solution to the SDE.

Let $v(t, X_t) = 1$ and $b_t = 0$. Then the SDE reduces to $dX_t = (aX_t)dt + dW_t, X_0 = x_0$.

Let $f(t, X_t) = \exp\{-at\}X_t$.

$$f_t = -a \exp\{-at\}X_t, f_X = \exp\{-at\}, f_{XX} = 0$$

. By theorem 3.1.2, we have

$$\exp\{-at\}X_t = x_0 + \int_0^t \{-a \exp\{-as\}X_s + a \exp\{-as\}X_s\} ds + \int_0^t \{\exp\{-as\}\} dW_s$$

.

$$X_t = x_0 \exp\{-at\} + \int_0^t \{\exp\{a(t-s)\}\} dW_s$$

is the solution.

Example 3.2.3

Consider the SDE

$$dX_t = -\text{sgn}(X_t)dW_t, X_0 = x_0,$$

where,

$$\text{sgn}(X_t) = \begin{cases} 1, & \text{if } X_t > 0. \\ -1, & \text{if } X_t < 0. \end{cases} \quad \text{The above SDE has no strong or weak solution. The Integral form of the SDE is } X_t = x_0 - \int_0^t \text{sgn}(X_s)ds .$$

We show that there exists no solution to the above SDE by contradiction. If possible, let us suppose it has a solution, say Y such that $\exists a > 0, t > 0, Y_t = a$.

Define τ and σ as

$$\tau = \inf \{t \geq 0 | Y_t = a\} \text{ and } \sigma = \sup \{t \leq \tau | Y_t = 0\}.$$

Using the integral form of the SDE, we get $Y_\tau = Y_\sigma = (\sigma - \tau)$. Hence, $Y \leq 0$. Similarly, we can show that $Y \geq 0$ which would imply that $Y = 0$, which is not a solution of the SDE. Hence we arrive at a contradiction.

Thus the SDE has no strong or weak solutions.

Chapter 4

Numerical Solutions of SDEs

Most problems which require a mathematical solution, be it in Finance, Physics, Biology or other disciplines, are stochastic but are usually solved using deterministic models. Rather than trying to formulate the exact phenomena, we could give probabilities to the occurrence of the event but for the sake of convenience in mathematically formulating the problem, we avoid this approach. This renders the problem simpler to deal with. However, with the deterministic approach, the accumulation of the errors due to stochastic effects has little or no effect over a short time frame but may have an adverse effect on the results over a longer time horizon. To overcome this problem, its important to implement a method to consider the stochastic effects. However, by the central limit theorem, we know that errors are normally distributed for simpler cases where the error is due to error in measurement. Stochastic models therefore, are better suited to capture real phenomena.

Stochastic Differential Equations are difficult to solve explicitly theoretically and hence have to be solved numerically for practical applications. There are various methods of solving them numerically, most rely on computer programs and with the current capabilities of the modern computers, this task is not as arduous anymore. These methods mostly revolve around implementing simulations of a large number of sample paths of the solutions of the SDEs and comparing their statistical features with the desired solutions.

One of the numerical methods involves investigating the random systems using Monte-Carlo Simulations introduced by Boyce(1978) but this did not exploit the special structure of these equations with the drift and diffusion coefficients. Another approach, introduced by Kushner and Dupuis, is the discretization of time and space variables which results in the approximating processes being finite state Markov chains which is preferable for low

dimension problems. This can be implemented on a computer using their transition matrices but it becomes too cumbersome to deal with for higher order problems due to the increase in the order of the transition matrix with increase in the dimensions of the problem and leads to loss of valuable computing resources in repeatedly evaluating the transition matrices, not all of which is required.

The preferred approach of solving SDEs is by simulating the sample solution paths of finite discretized time approximations and using statistical methods to analyze how close the approximation is to the exact solution. This method seems to be the most popular since its parsimonious in terms of both time and computing resources as the required memory increases polynomially with the increase in the dimension of the problem.

4.1 Monte-Carlo Simulations

Monte Carlo simulation methods are computational algorithms used for generating a large number of random sample paths to a particular quantitative problem to statistically evaluate the solutions. This approach is used extensively in many areas which are difficult to solve explicitly. In mathematical finance, it proves to be an indispensable tool for stress-testing and scenario analysis of many models used in finance, for instance testing volatility forecasting models. Its a widely known fact that Monte Carlo Simulations play an important role in finance in risk management and other areas. (For further reading, see [9], [10] , [16] and [19])

Consider the Stochastic Differential Equation

$$dX_t = u(t, X_t)dt + v(t, X_t)dW_t \quad (4.1)$$

where W_t is the Wiener process, $u(t, X_t)$ and $v(t, X_t)$ are the drift and diffusion coefficients on the time interval $[0, T]$. Let $0 = t_0 < t_1 < \dots < t_N = T$ be a finite discretisation of the time interval $[0, T]$, where N is the number of time steps, and is usually taken to be equidistant for ease in computation. The step size $\delta_n = h = (T - t_0)/N$ for an integer N large enough such that $h \in (0, 1)$.

To simulate a trajectory using the Monte Carlo Method, we start with an initial value which may be chosen randomly or for a given known initial value. We then generate an approximation sequence using the following iterative

scheme for (4.1)

$$Y_{n+1} = Y_n + (t_{n+1} - t_n)u(t_n, Y_n) + (W_{t_{n+1}} - W_{t_n})v(t_n, Y_n),$$

for $n=0,1,2,\dots,N-1$.

or

$$Y_{n+1} = Y_n + \delta_n u(t_n, Y_n) + (\delta W_{t_n})v(t_n, Y_n) \quad (4.2)$$

We know W is the Wiener process and hence the Wiener increments, δW_{t_n} are normally distributed and have mean 0 and variance t_n . Thus, we can write $\delta W_{t_n} = Z_{t_{i+1}} \sqrt{\delta t_n}$ where $Z_{t_{i+1}}$ is an i.i.d standard normal random variable. Hence (4.2) can be re-written as

$$Y_{n+1} = Y_n + \delta_n u(t_n, Y_n) + (Z_{t_{i+1}} \sqrt{(\delta t_n)})v(t_n, Y_n)$$

The use of Monte Carlo Simulations for generating sample paths of Stochastic Differential Equations of the form given by (4.1) is illustrated in Figure 4.1 through a MATLAB code for an SDE $dX_t = 1.5dt + 2dW_t$. In this example, the drift and diffusion coefficients have been taken as constants for the sake of convenience but could also be done for functions of X_t .

Figure 4.1 : Monte Carlo Simulation for the SDE $dX_t = 1.5dt + 2dW_t$

The following code produces the above Figure 4.1.

```
% Single Monte-carlo Simulation
N=1000; T = 1; h = T/N; Z = randn(N,1); X(1,1) = randn; mu1 = 1.5; sigma1= 2;
Z(1,1)= 0;

for i=1:N-1
    X(i+1,1) = X(i,1)+mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h) ;
end

t = linspace(0,N,N);
figure(fig_no)
title('Monte-Carlo Simulation ');
xlabel('Time'); ylabel('Sample Path');grid on; hold on;
plot(t' , X );
set(gcf,'color','white');
fig_no = fig_no + 1;
%%
```

The following figure shows the Monte Carlo simulations for 100 trajectories for the same SDE as in the previous example. With increase in the number of simulations, the results are seen to converge to the real solution.

Figure 4.2 : 100 trajectories of Monte Carlo Simulation for the SDE $dX_t = 1.5dt + 2dW_t$ (See Appendix M3)

We see that with decrease in the step size, h , the error term approaches 0 i.e. the approximation converges to the real solution. This is illustrated through Figure 4.3.

Figure 4.3 : Monte Carlo Simulation results for varying step sizes for the SDE $dX_t = 1.5dt + 2dW_t$ (See Appendix M4)

4.2 The Euler Approximation

The simplest time discrete and intuitive approximation is the Euler Approximation and is defined analogously to that of the Ordinary Differential Equation but with some changes due to the additional noise term in SDEs. (See [10], [20] and [21])

Consider an Itô Process $X = (X_t)$ represented by

$$X_t = X_0 + \int_0^t u(s, X_s)ds + \int_0^t v(s, X_s)dW_s$$

where W_s is the Wiener Process, $u(s, X_s)$ represents the slowly moving continuous drift and $v(s, X_s)$ represents the continuous diffusion component for $t \in [0, T]$.

Its represented as a Stochastic Differential equation by (4.1).

Let the initial value be $X_{t_0}(\omega) = X_0(\omega)$ and $0 = t_0 < t_1 < \dots < t_N = T$ be a finite discretisation of the time interval $[0, T]$.

To simulate a trajectory of the Euler approximation for a given time discretization, we simply start with the initial value, $Y_0 = X_0$. We then proceed iteratively to generate the approximation sequence by the following equation

$$Y_{n+1} = Y_n + (t_{n+1} - t_n)u(t_n, Y_n) + (W_{t_{n+1}} - W_{t_n})v(t_n, Y_n)$$

or

$$Y_{n+1} = Y_n + (\delta_n)u(t_n, Y_n)Y_n + (\delta W_{t_n})v(t_n, Y_n)Y_n \quad (4.3)$$

for $n = 0, 1, 2, \dots, N - 1$ with drift and diffusion coefficients u and v and $Y_n = Y(t_n)$ is the n th value of the approximation sequence.

As explained in section (4.1), we can write $\delta W_{t_n} = Z_{t_{i+1}} \sqrt{\delta t_n}$ where $Z_{t_{i+1}}$ is an i.i.d standard normal random variable. Thus, (4.3) can be re-written as

$$Y_{n+1} = Y_n + (\delta_n)u(t_n, Y_n)Y_n + (Z_{t_{i+1}} \sqrt{\delta t_n})v(t_n, Y_n)Y_n$$

One needs to exercise caution while defining the above iterative equation for the approximation sequence to ensure that it makes sense as increments of the Wiener Process can have very large jumps with either sign but in reality this is not very likely to happen. This poses a serious problem for non-linear equations.

The maximum step size is given by $h = \delta = \text{Max}_n \delta_n$, where $\delta_n = t_{n+1} - t_n$ is the n th time increment. The discretization or partition of time can be equally spaced or may be varying but its easier to work with equidistant times given by

$t_n = t_0 + nh$, where $h = \delta_n = (T - t_0)/N$, for an integer N large enough such that $h \in (0, 1)$.

After the approximation sequence is obtained from equation (4.3), it remains to determine how good the approximation is, that is to say, how close the sample path is to the exact solution. There are essentially two ways of doing this and depends greatly on the objective of the simulation.

The Strong Convergence Criterion

This method is implemented for evaluating the performance of the approximation whenever a good path wise approximation is required, for instance, in direct simulations, filtering problems or testing statistical estimators.

Definition 4.2.1 : Convergence in the strong sense with order γ

A sample path of the approximation, Y_n given by (4.3) is said to converge in the strong sense with order $\gamma \in (0, \infty]$ if $\exists$ constants $K < \infty$ and δ_0 such that $E[|X_T - Y_N|] \leq Kh^\gamma$ where $h \in (0, \delta_0)$ is the maximum step size.

However, for all practical purposes the quality of the approximation is judged

by the absolute error term which gives the expectation of the absolute difference at the final state of time of approximation sequence, Y_N and the Itô process, X_T given by,
 $\epsilon(h) = E[|X_T - Y_N|]$, and estimated from the root-mean square error using the Lyapunov Inequality.²

In the case when the diffusion coefficient is non-existent i.e. in the absence of the noise, the absolute error criterion collapses into the absolute error criterion for ordinary differential equations, the global truncation error.

We see that with decreasing step size, h , the error term, $\epsilon(h)$ approaches 0. For equidistant step sizes, h approaches 0 as N approaches ∞ and consequently the error, $\epsilon(h)$ approaches 0. As mentioned before in the chapter, a consequence of the Central Limit theorem is that the error term, also a random variable, follows the Gaussian distribution as N approaches ∞ and converges in distribution to the deterministic error. This is illustrated by Figure 4.6.

The Weak Convergence Criterion

The strong convergence Criterion is not always required and in many cases, depending on the aim of the simulation, the Weak Convergence Criterion is actually preferred, like when approximating some function value equal to that of the Itô process at the final time T , where the function could be the expectation, probability, a polynomial, etc.

Definition 4.2.2 : Convergence in the strong sense with order γ

A sample path of the approximation, Y_n given by (4.3) is said to converge in the weak sense with order $\beta \in (0, \infty]$ if for any polynomial g , $\exists$ a constant $K < \infty, \delta_0$ such that $|E[g(X_t)] - E[g(Y_N)]| \leq Kh^\beta$ for any time discretization with maximum step size $h \in (0, \delta_0)$

Example 4.2.1

For an illustration, consider the following equation -

$$dS_t = S_t(\mu + \sigma dW_t), \quad (4.4)$$

where S_t represents the stock price, μ and σ represents the drift and the volatility on the financial markets respectively and W_t is the usual Wiener

²Lyapunov Inequality : $E[|Y|^s]^{1/s} \leq E[|Y|^t]^{1/t}$ for $0 < s < t$. From this inequality we get, for $s = 1 < t = 2$, $\epsilon(h) = E[|X_T - Y_N|] \leq (E[|X_T - Y_N|^2])^{1/2}$

Process or Brownian motion.

Comparing the above equation with equation (4.1),
 $u(t, S_t) = \mu S_t$, $v(t, S_t) = \sigma S_t$.

Using Itô's Lemma, we know that SDE (4.4) has the solution as done in chapter 3

$$S_t = S_0 \exp\left\{\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right\} \quad (4.5)$$

To simulate a trajectory of equation (4.4), we start with the initial value $Y_0 = S_0$, which could be randomly chosen. The iterative equation (4.3) then can be written as

$$Y_{n+1} = Y_n + \mu Y_n \delta_n + \sigma Y_n \delta W_n, \quad (4.6)$$

where $\delta W_n = W_{t_{n+1}} - W_{t_n}$ is the increment in the Wiener Process.

From the definition of Brownian motion, we know that the increments of W_t are Gaussian, i.e. Mean = $E[\delta W_n] = 0$ and Variance = $E[(\delta W_n - E[\delta W_n])^2] = \delta_n$ i.e. $\delta W_n \sim N(0, \delta_n)$.

Hence we can use a pseudo-random number generator, available in all programming languages to generate independent Gaussian increments with mean 0 and variance δ_n to construct the Wiener Process and for known values of μ and σ , we can then generate the approximation sequence. The approximation sequence or the sample path of the resulting Itô Process thus obtained is also continuous but non-differentiable. This describes the procedure for implementation of the Euler Approximation method.

For comparison of the approximation sequence thus obtained with the corresponding values of the exact solution i.e. equation (4.5) , we can compare the values of the approximation sequence with the following- $S_{t_n} = S_0 \exp\left\{\left(\mu - \frac{\sigma^2}{2}\right)t_n + \sigma \sum_{i=1}^n \delta W_{i-1}\right\}$.

Figure 4.4 illustrates the above example for the SDE $dX_t = 1.5dt + 2dW_t$ with the MATLAB code.

1 Figure 4.4 : Sample Path by Euler Approximation for the SDE $dX_t = 1.5dt + 2dW_t$

The following code shows how to simulate a sample path by the Euler Approximation.

```
% Single Sample path for Euler Approximation
N=1000; T = 1; h = T/N; Z = randn(N,1); Y(1,1) = randn; mu1 = 1.5; sigma1= 2;
Z(1,1)= 0;

for i=1:N-1
    Y(i+1,1) = Y(i,1)+ Y(i,1)* mu1 *h + (Z(i+1,1)) * sigma1 * Y(i,1)*sqrt(h);
end

t = linspace(0,N,N);
figure(fig_no)
title('Euler Approximation');
xlabel('Time'); ylabel('Sample Path');grid on; hold on;
plot(t' , Y );
set(gcf,'color','white');
fig_no = fig_no + 1;
```

Clearly, the sample paths obtained from the Euler Approximation are dense as shown in Figure 4.5, in fact more than that for the Monte Carlo simulations, which suggests that with increase in the number of simulations, the sample paths converge to the real solution.

1 Figure 4.5 : 100 Sample Paths by Euler Approximation (See Appendix M5)

1 Figure 4.6 : Euler Approximation for varying step sizes (See Appendix M6)

Despite simulations using time-discrete approximations being the most preferred approach for solving SDEs, simulations of SDEs using this approach sometimes run into problems. The following example illustrates this.

Example 4.2.2

Consider the SDE - $dX_t = u(t, X_t)dt + \sqrt{2X_t}dW_t$. (See [10]) since $v(t, X_t) = \sqrt{2X_t}$ is not a Lipschitz function ³ since as X_t approaches 0, the function v becomes very steep and consequently $\frac{|\sqrt{2X_t}|}{t} \rightarrow \infty$, its

³The function $V(t, X_t)$ will be called Lipschitz continuous if $\exists$ a constant $K \geq 0$ such that $|V(t, X_t) - V(s, X_s)| \leq K|t - s|$ for all $s, t \in \mathbb{R}$

derivative approaches ∞ . Also, by theorem 3.2.1, the existence of a solution is not guaranteed as the Lipschitz condition fails.

$v(t, X_t) = (X_t)^{3/2} \sin(\frac{1}{X_t})$, $X_t \neq 0$, $X_t = 0$ defined on $[0, 1]$ is also another example which is not locally Lipschitz as its derivative is not bounded. These kind of SDEs are hence not very tractable even for numerical methods.

4.3 Possible Application in Finance

There is no dearth of forecasting volatility models like GARCH, EWMA, BEKK, etc and its important to test these models before they are used by financial institutions and choosing the best model among the ones available for forecasting risk is just as important. This end is achieved by back testing, a procedure used to check the accuracy of the model by testing the model with historical data. However, historical data is not available very easily. Other methods employed are stress testing, which involves creating artificial market outcomes to gauge the performance of the model and scenario analysis, where models are tested for extreme market outcomes and scenarios which are not captured well by historical data. Since we know that the stock prices

denoted by S_t follows the Geometric Brownian motion, $dS_t = \sigma S_t dt + S_t dW_t$, which was also the underlying assumption of the Black-Scholes model for pricing options, where σ represents the volatility, we could obtain sample paths for S_t using Monte Carlo simulations and/or Euler Approximation to compare with the actual path where σ or the time varying volatilities would be estimated by a model, say EWMA or GARCH. In this paper, we see the implementation of this approach but only for the EWMA and for a single asset though the same could be achieved for multiple assets or portfolios. This approach may offer some insights about the performance of these models and could be employed for a collection of all the models available or to be tested. Exponentially Weighted Moving Average or EWMA is a volatility forecast-

ing model expressed as a weighted sum of the previous day's volatility and squared returns where the weights add up to 1. Its defined as

$$\sigma_t^2 = (1 - \lambda)r_{t-1}^2 + \lambda\sigma_{t-1}^2$$

, where σ_t represents conditional volatility at time t given the squared returns

at time $t-1$, r_{t-1} and conditional volatility, σ_{t-1} at time $t-1$ and $0 < \lambda < 1$ denotes the decay factor and is taken to be 0.94. (See [14])

We have downloaded the closing prices for two companies, namely Citigroup and Microsoft for the past 10 years. However, we only use the data for Citigroup in this paper. The EWMA model is used to obtain the time varying volatilities which is used to simulate paths for S_t in the geometric Brownian motion model. The MATLAB code for the above explained approach is given briefly in Appendix M8.

Chapter 5

Applications

The use of Mathematics in the Social Sciences or more particularly in Economics is often debated by Economists and Mathematicians alike. Some have gone as far as saying that the recent financial crisis has been exacerbated as a result of using all the complicated Mathematics and relaxing the assumptions underlying the results without understanding the implications in their entirety. Others have argued that its a result of using outdated Mathematical tools when the present situation calls for much more sophisticated Mathematical models. But nonetheless, Mathematics has been used in the past and continues to be used in Economics and as expected, it never fails to surprise and very often yields very interesting results, even in Economics.

In this chapter, we briefly introduce an interesting problem in Economics modeled by a Stochastic Differential Delay Equation (SDDE). However, being an optimization problem, its not explicitly solved here but only reflects the beauty of the model and to illustrate how a practical problem can be modeled mathematically from scratch.

In 1928, Mathematician and Economist, F.P.Ramsey in his seminal paper[11], A Mathematical Theory of Saving, established a model of economic capital and labour through an Ordinary Differential Equation. The paper revolves around the question - "How much of its income should a nation save?", that is, to maximize the future utility such that the consumption is optimal. We first describe this model followed by the stochastic model of this problem in the next section.

5.1 Ordinary Differential Equation Model

Let us first introduce some notations.

$C(t)$ represents the capital available to Society at time t

$X(t)$ represents the total rate of consumption or expenditure at time t

$L(t)$ represents the Labour rate of our Society at time t

$P(t)$ represents the production rate at time t

$U(X)$ represents the total rate of utility of a rate of consumption, X

$V(L)$ represents the total rate of dis-utility of a rate of labour, L

The Corresponding marginal rates are given by

$$u(X) = \frac{dU(X)}{dX}; v(L) = \frac{V(L)}{dL}$$

For a fixed given level of capital, its easy to see that the marginal utility rate of consumption, $u(X)$ never increases and the marginal utility rate of labour, $v(L)$ is never decreasing.

The first fundamental equation is

$$\frac{dC(t)}{dt} = P(t) - X(t) \quad (5.1)$$

(5.1) expresses the relationship that savings plus the consumption should be equal to the income or the production. The production, P is actually a function of capital, C and Labour, L but both are functions of time hence $P = P(t)$. By the Cobb-Douglas formula, we have

$$P(t) = AC^\alpha(t)L^\beta(t)$$

where A, α, β are positive constants representing the total factor productivity and output elasticities of capital and labour respectively and can be obtained statistically. We further assume, $\alpha = 1, \beta = 1$, so P will be linear in L and C . Also, its assumed that over short time intervals, labour is constant, so $L(t) = L_0$. Thus, we have $P(t) = AC(t)L(t) = (AL_0)C(t) = MC(t)$ where

$M = AL_0$ is a constant.

In order to answer the question posed by Ramsey, one needs to find the optimal consumption rate such that society attains the maximum possible enjoyment from the income and can then save the remaining for future. $U(X) - V(L)$ represents society's total enjoyment per unit time for the given fixed level of capital C . Mathematically, this translates to maximising $(U(X) - V(L))$ subject to the condition that Society's consumption is what can be produced with labour, L and capital, C .

The resulting rate of enjoyment, $U(X) - V(L)$ is a function of capital, C . Since with increase in capital, enjoyment, $(U(X) - V(L))$ increases but up to a certain limit beyond which any additional capital cannot give any additional enjoyment. The following two cases give concrete form to the above assertion.

Case I : The increase in enjoyment with increasing capital may stop after a certain limit due to the following reasons

- (i) The enjoyment may have attained a maximum, so with increase in capital, C , there may not be any more need for more enjoyment or income
- (ii) A further increase in capital may not enable us to increase society's income or enjoyment

Case II : The increase in capital may not lead to a stop in the rate of enjoyment

In the case that the rate of enjoyment never stops with increase in capital, there could be two situations

- (i) Rate of enjoyment approaches ∞ . This case can be rejected immediately since society or people cannot possibly derive more than a certain finite amount of enjoyment out of economic reasons alone. This finite rate of enjoyment is called a maximum conceivable rate of enjoyment.
- (ii) The rate of enjoyment approaches a certain finite amount asymptotically, which is called the maximum obtainable rate of enjoyment. This may or may not equal the maximum conceivable rate.

For brevity, the maximum obtainable rate of enjoyment is called bliss and denoted by B .

Thus, in each case, Society has to save enough to either approach bliss, B indefinitely or attains bliss after a finite time.

Now, the marginal dis-utility of labour at any time t , $v(L)$ equals the product of marginal efficiency of labour, $\frac{\partial P}{\partial L}$ and the marginal utility of consumption

, $u(X)$ i.e.

$$v(L) = \frac{\delta P}{\delta L} u(X) \quad (5.2)$$

Also, the difference between the gains drawn from an increase, ΔX in the consumption at time t and that drawn by postponing it for an infinitesimal period, Δt is proportional to that rate of interest earned by waiting, $\frac{\delta P}{\delta C}$. Thus we have the following equation

$$\frac{u(X(t + \Delta t)) - u(X(t))}{(\Delta t)} = \frac{\delta P}{\delta C} u(X(t + \Delta t))$$

As $\delta t \rightarrow 0$, we have ,

$$\frac{du(X(t))}{dt} = -\frac{\delta P}{\delta C} u(X(t)) \quad (5.3)$$

i.e. the marginal utility of consumption, $u(X)$ falls at a rate proportional to the rate of interest.

We have at our hand three equations, (5.1), (5.2) and (5.3) to work with and the initial conditions : C_0 is the initial capital with which a nation starts at time $t = 0$ and the other conditions provided by the behaviour of functions X , L and C as time $t \rightarrow \infty$ which are all functions of time, t only.

By using the chain rule,

$$\begin{aligned} & \frac{d}{dX}(u(X)P(C, L)) \\ &= \frac{du(X)}{dX}P(L, C) + \{u(X)\frac{\delta P(L, C)}{\delta L}\}\frac{\delta L}{\delta X} + \{u(X)\frac{\delta P(L, C)}{\delta C}\}\frac{dC}{dt}\frac{dt}{dX} \end{aligned}$$

Substituting from (5.2) and (5.3) respectively in above equation, we have

$$= \frac{du(X)}{dX}P(L, C) + v(L)\frac{\delta L}{\delta X} - \frac{du(X)}{dt}\frac{dC}{dt}\frac{dt}{dX}$$

Now substituting for $\frac{dC}{dt}$ from (5.1) in above,

$$\begin{aligned} &= \frac{du(X)}{dX}P(L, C) + v(L)\frac{\delta L}{\delta X} + \frac{du(X)}{dt}\frac{dt}{dX}\{P(L, C) - X\} \\ &= x\frac{du(X)}{dX} + v(L)\frac{\delta L}{\delta X} \end{aligned}$$

Thus, we have

$$\frac{d}{dX}(u(X)P(C, L)) = x \frac{du(X)}{dX} + v(L) \frac{\delta L}{\delta X}$$

Integrating above equation by parts,

$$u(X)P(L, C) = Xu(X) - U(X) + V(L) + K$$

, where K is a constant. i.e.

$$u(X)(P(L, C) - X) = -U(X) + V(L) + K$$

$$u(X) \frac{dC}{dt} = K - (U(X) - V(L))$$

Thus, we have

$$\frac{dC}{dt} = \frac{K - (U(X) - V(L))}{u(X)} \quad (5.4)$$

Now we have to find the constant, K and this is achieved by a slightly different approach.

Consider $\int_0^\infty (B - U(X) + V(L))dt$, which represents the difference between the maximum obtainable rate of enjoyment or Bliss, B and the rate of enjoyment, $U(X) - V(L)$. The aim is to minimise the integral so that bliss can be attained.

We first change the independent variable from t to C and the limits vary from $t = 0 \Rightarrow C = C_0$ to $t \rightarrow \infty \Rightarrow C \rightarrow \infty$

As capital C increases with time till $(B - U(X) + V(L)) = 0$ and is never ∞ but the minimum capital with which bliss, B can be attained. So we can the integral as ,

$$\int_0^\infty \frac{(B - U(X) + V(L))}{\frac{dC}{dt}} dC$$

Using (5.1) again, we have

$$\int_0^\infty \frac{(B - U(X) + V(L))}{P(L, C) - X} dC$$

Now minimising the above integral is equivalent to minimising the integrand. We differentiate with respect to X and equate to zero, using the simple procedure of calculus.

i.e.

$$\frac{-(P(L, C) - X)(\frac{dU(X)}{dX}) + (B - U(X) + V(L))}{(B - U(X) + V(L))^2} = 0$$

Substituting from (5.1) and marginal utility of consumption,

$$\frac{-(\frac{dC}{dt})u(X) + (B - U(X) + V(L))}{(B - U(X) + V(L))^2} = 0$$

Thus, we get from above,

$$\frac{dC}{dt} = \frac{B - (U(X) - V(L))}{u(X)}$$

Comparing the above equation with (5.4), we get $K = B$

5.2 Stochastic Differential Equation Model

In addition to the assumptions made above for the ODE model of the problem, we assume that the production rate is subject to "noise", i.e. the equation $P(t) = MC(t)$ in the ODE problem now becomes, $P(t) = MC(t) + \text{"noise"}$. This is due to [1]. (Also see [5])

Then (5.1) now becomes,

$$\frac{dC}{dt} = MC(t) + \text{"noise"}$$

which can be written in differential form by Itô's representation by,

$$dC(t) = \{MC(t) - X(t)\}dt + \sigma(C(t))dW(t) \quad (5.5)$$

where $W(t)$ is the standard Wiener Process and $\sigma(C(t))$ is a given real function characteristic of the noise.

A tacit assumption made by the Rasmeys model, is that there's instant transformation of investments but in practice this does not hold. A more reasonable assumption is that it takes some time before the investment transforms into working capital, so the rate of change of capital at time t depends on the investment made at an earlier time, $(t-T)$, where T is the time taken for the creation of working capital[11]. This is accounted for by re-writing (5.4) as follows

$$dC(t) = \{MC(t - T) - X(t)\}dt + \sigma(C(t - T))dW(t) \quad (5.6)$$

called the Stochastic Delay Differential Equation.

A further assumption made is that the consumption rate, $X(t)$ depends on the capital available, which is a realistic assumption, and is represented by $X(t)u(C(t))$ where u is the control. For $T = 1$, i.e by normalising the delay, we have

$$dC(t) = \{MC(t - 1) - X(t)u(C(t))\}dt + \sigma(C(t - 1))dW(t) \quad (5.7)$$

The initial condition is given by : the initial investment is represented by a continuous function, $C(t) = \phi(t)$ where $t \in [-1, 0]$.

Equation (5.7) and the above condition presents the Stochastic version of the Ramsey model.

Although, the solution of (5.7) is beyond the scope of this paper at present, it does serve as an interesting problem for consideration in the future.

5.3 Application in Biology

The Ornstein-Uhlenbeck Process is a stochastic process which progresses to a constant or its long term mean with time but is subject to randomness or noise continuously and plays an important role in many disciplines such as mathematical finance, statistical mechanics, biology and elsewhere. For example, consider a model in physics characterised by the above process where a particle with velocity $V(t)$ at time t moving through a fluid or gas will have a frictional drag and a noise term due to the collision of molecules around the particle.

The Ornstein-Uhlenbeck process is also exhibited by the membrane potential of a neuron. The plasma membrane around the animal cells contain lipid bilayers of various types of proteins, the interaction between these types of transmembrane proteins and the membrane leads to the membrane potential, which is the difference between the electrical impulses between the interior and exterior of the biological cell. On one hand, the membrane potential is attracted towards an equilibrium value determined by the potentials of the various ions at various concentration levels within the cell and at the space between the tissues, whereas on the other hand, the membrane potential is subject to randomness due to the electrical impulses arising from the surrounding network. This is modelled by the SDE given below. (See [25])

Consider the Stochastic Differential Equation given by

$$dX_t = -\frac{(X_t - \alpha)}{\tau} dt + \sigma dW_t, X(0) = X_0$$

is the initial condition and τ and α are constants. $u(t, X_t) = -\frac{(X_t - \alpha)}{\tau}$ and $v(t, X_t) = \sigma$

Using theorem 3.1.2 and letting $f(t, X_t) = \exp\{\frac{t}{\tau}\}X_t$, and substituting

$$f_t(t, X_t) = \frac{1}{\tau} \exp\{\frac{t}{\tau}\}X_t, f_X(t, X_t) = \exp\{\frac{t}{\tau}\}, f_{XX}(t, X_t) = 0 \text{ in (3.1).}$$

We have

$$f(t, X_t) = X_0 + \int_0^t \left\{ \frac{1}{\tau} \exp\left\{\frac{s}{\tau}\right\} X_s - \frac{(X_s - \alpha)}{\tau} \exp\left\{\frac{s}{\tau}\right\} \right\} ds + \int_0^t \sigma \exp\left\{\frac{t}{\tau}\right\} dW_t$$

$$X_t = \exp\left\{\frac{-t}{\tau}\right\} X_0 + \frac{\alpha}{\tau} \exp\left\{\frac{-t}{\tau}\right\} \int_0^t \exp\left\{\frac{s}{\tau}\right\} ds + \exp\left\{\frac{-t}{\tau}\right\} \int_0^t \sigma \exp\left\{\frac{s}{\tau}\right\} dW_s$$

$$X_t = \exp\left\{\frac{-t}{\tau}\right\} X_0 + \frac{\alpha}{\tau} \exp\left\{\frac{-t}{\tau}\right\} (\tau \exp\left\{\frac{t}{\tau}\right\} - 1) + \exp\left\{\frac{-t}{\tau}\right\} \int_0^t \sigma \exp\left\{\frac{s}{\tau}\right\} dW_s$$

$$X_t = \exp\left\{\frac{-t}{\tau}\right\} X_0 + \alpha(1 - \exp\left\{\frac{-t}{\tau}\right\}) + \exp\left\{\frac{-t}{\tau}\right\} \int_0^t \sigma \exp\left\{\frac{s}{\tau}\right\} dW_s$$

is the required solution of the SDE.

Chapter 6

Conclusion

Stochastic Calculus not only has myriad applications but is a very rich area of pure Mathematics in itself and this paper . To begin with, this paper reviews some basic measure theoretic concepts, which are very closely linked to stochastic calculus, for instance in defining the Itô integral, and tries to draw similarities and analogous interpretations wherever possible. Thereafter, Stochastic processes and some important results which are used throughout this paper are introduced followed by the construction of the Itô integral. For a geometric interpretation of some concepts like Brownian motion, although hard to visualise, and to provide a clearer picture of theoretical definitions, some MATLAB programs have been implemented to generate the graphs.

Solving Stochastic Differential Equations is a far more challenging task than solving Ordinary Differential Equations due to the additional "noise" term and Itô's formula serves as a powerful tool in overcoming this problem. The existence and uniqueness theorem enables us to identify the Stochastic Differential Equations, satisfying the Lipschitz and growth conditions, for which a solution exists. Some examples are worked out to illustrate this and the two types of solutions, weak and strong solutions.

However, many Stochastic Differential Equations may be difficult to solve explicitly but there are various numerical methods, such as Monte-Carlo Simulations, Euler Approximation, Milstein Scheme, etc to approximate the solution of a SDE. Only two of these numerical methods are discussed in some detail in this paper, namely Monte-Carlo and Euler simulations, along with their implementation in MATLAB. Moreover, an attempt is made to classify some volatility forecasting models, such as EWMA (Exponentially Weighted Moving Average model) using the Geometric Brownian Motion model. The

stock prices of Citigroup for the past 10 years is downloaded and is used to forecast the time varying volatilities and is used in the Geometric Brownian motion model and a sample path is obtained using MC or Euler Simulation. This could then be compared with the real prices and could be a way of evaluating the performance of these forecasting models.

In the last chapter, an important paper by F.P.Ramsey which addresses a very fundamental economic question is explained at some length to construct the Ordinary Differential Equation model followed by the stochastic model of this problem which leads to the Stochastic Differential Delay Equation. Thereafter, a model in Biology is described in which the membrane potential of neurons is seen to exhibit the Ornstein-Uhlenbeck process.

Further, I am including some suggestions for possible extension of this paper. Firstly, the stochastic version of Ramsey's model presents an interesting optimisation problems for consideration in the future and further study of Stochastic Differential Delay Equations. Secondly, with the multitude of volatility forecasting models available at the hands of decision makers, it may be an overwhelming decision to choose one model over the other, and the method proposed in chapter 4 may be provide some respite.

I hope this paper proves to be a helpful source of reference in that it provides a succinct description of the theory of stochastic calculus with some graphs and worked out examples, and at the same time draws similarities with measure theory to give the reader, irrespective of the reader's familiarity with measure theory, a better understanding of the subject. It also highlights some important applications.

Appendix A

MATLAB Codes

All the codes in this paper have been written without keeping performance issues in mind since the computations carried out through these programs do not require very high computational power.

Some codes below generate figures which are not included in the chapters as they are only for feeding the curiosity. All the files, MATLAB script and function files as well as the excel data file, have to be kept in the same folder in the current directory in MATLAB to ensure smooth functioning of the codes.

A.1 M1 - Figure 2.1

```
%% One-Dimensional Brownian Motion

clc;clear all; clf;
N=1000;% the number of data points
seed = randn(N-1,1);
% creates (N-1) X 1 vector of pseudo-random
%variables drawn from the normal distribution
Time_vec = linspace(0,1,N);
brown = [0;cumsum(seed)]/sqrt(N);% creates 1-dim brownian motion

%% Plotting the Weierstrass function

a = 0.5; b = 3; kmax = 20;
c1(1:kmax+1) = a.^(0:kmax);
c2(1:kmax+1) = 2*pi*b.^(0:kmax);
```

```

x = linspace(0,1,1000);

% plotting one-dim BM and Weierstrass Function
clf;
fig_no          = 1;
figure(fig_no);
subplot(1,2,1)
title('Time evolution of one-dimensional brownian motion');
xlabel('Time'); ylabel('BM');grid on; hold on;
plot(Time_vec',brown,'LineStyle','-','LineWidth',1,'Color','blue');

subplot(1,2,2)
title('Weierstrass''s non-differentiable function');
xlabel(['a=' num2str(a), ', b=' num2str(b)]);
plot(x,w(x,c1,c2),'b-','LineWidth',2);
set(gca,'color','white');
grid on;
fig_no          = fig_no + 1;

figure(fig_no);
title('Distribution of the randomly generated numbers');
% The histogram gives the normal distribution but the figure is not included.
hist(brown,100);
fig_no          = fig_no + 1;

```

The following MATLAB function file is used to generate the graph of the Weierstrass Function used in the previous code.[]

```

function [ y ] = w( x, c1, c2 )
index = 1;
y = zeros(length(x),1);
for k = x
y(index) = sum(c1 .* cos(c2*k));
index = index + 1;
end

end

```

A.2 M2. Figure 2.2

```

%% Two - Dimensional Brownian Motion

```

```

N = 1000;
seed1 = randn(N,1);
seed2 = randn(N,1);
T = 10/N;
sigma = 1;

brown1 = zeros(N,1);% initialising the vectors
brown2 = zeros(N,1);

for i=1:N-1
    brown1(i+1,1) = brown1(i,1) + sqrt(T)*sigma*seed1(i,1);
    brown2(i+1,1) = brown2(i,1) + sqrt(T)*sigma*seed2(i,1);
end

figure(fig_no);
title('Time evolution of two-dimensional brownian motion');
xlabel('Time'); ylabel('BM');grid on; hold on;
plot(brown1,brown2,'LineStyle','-','LineWidth',1,'Color','blue');
fig_no      = fig_no + 1;

```

A.3 M3. Figure 4.2

```

%% MC Simulation for 100 trajectories
T=1; N = 1000; seed = randn; num_traj = 100;
h = T/N;% step size
Z = randn(N,100); mu1 = 1.5; sigma1= 2; X = zeros(N,num_traj);

Z(1,1:num_traj)= 0;

for j=1:num_traj
    X(1,j) = seed;
    for i=1:N-1

        X(i+1,j) = X(i,j) + mu1*h + sigma1*Z(i,j)*sqrt(h);
    end
end
t = linspace(0,N,N);

figure(fig_no)
title('100 trajectories of Monte-Carlo Simulation');

```

```

xlabel('Time'); ylabel('Sample Paths');grid on; hold on;
plot(t' , X(:,1:100));
set(gcf,'color','white');
fig_no = fig_no + 1 ;

```

A.4 M4. Figure 4.3

The following code produces Monte Carlo simulation results for varying step sizes.

```

%% Multiple MC simulations for varying number of simulations
seed = randn; mu1 = 1.5; sigma1= 2; Z(1,1)= 0;
%% for N=10;
N=10; T = 1; h = T/N; Z = randn(N,1); X1 = zeros(N,1);
X1(1,1) = seed;

for i=1:N-1
    X1(i+1,1) = X1(i,1)+ mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h);
end
t1 = linspace(0,N,N);
%% for N=100;

N=100; T = 1; h = T/N; Z = randn(N,1); X2 = zeros(N,1);
X2(1,1) = seed;

for i=1:N-1
    X2(i+1,1) = X2(i,1)+ mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h);
end
t2 = linspace(0,N,N);
%% for N=1000;

N=1000; T = 1; h = T/N; Z = randn(N,1); X3 = zeros(N,1);
X3(1,1) = seed;

for i=1:N-1
    X3(i+1,1) = X3(i,1)+ mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h);
end
t3 = linspace(0,N,N);
%% for N = 10000;

N=10000; T = 1; h = T/N; Z = randn(N,1); X4 = zeros(N,1);

```

```

X4(1,1) = seed;

for i=1:N-1
X4(i+1,1) = X4(i,1)+ mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h);
end
t4 = linspace(0,N,N);

%% Plotting for diff values of h

figure(fig_no)
title('Monte-Carlo Simulation for varying step sizes ');
subplot(4,1,1)
xlabel('Step Size , h = 0.1'); ylabel('Sample Path');grid on; hold on;
plot(t1' , X1 );
subplot(4,1,2)
xlabel('Step Size , h = 0.01'); ylabel('Sample Path');grid on; hold on;
plot(t2' , X2 );
subplot(4,1,3)
xlabel('Step Size , h = 0.001'); ylabel('Sample Path');grid on; hold on;
plot(t3' , X3 );
subplot(4,1,4)
xlabel('Step Size , h = 0.0001'); ylabel('Sample Path');grid on; hold on;
plot(t4' , X4 );

set(gcf,'color','white');
fig_no = fig_no + 1;

```

A.5 M5. Figure 4.5

100 trajectories by Euler Simulation

```

% 100 Sample Paths using Euler Approximation
T=1; N = 1000; seed = randn; num_traj = 100;
h = T/N;% step size
Z = randn(N,num_traj); mu1 = 1.5; sigma1= 2; Y = zeros(N,num_traj);

Z(1,1:num_traj)= 0;

for j=1:num_traj
    Y(1,j) = seed;
    for i=1:N-1

```

```

Y(i+1,j) = Y(i,j)+ Y(i,j)* mu1 *h  + (Z(i+1,j)) * sigma1 * Y(i,j)* sqrt(h);
end
end
t = linspace(0,N,N);

figure(fig_no)
title('100 Sample Paths using Euler Approximation');
xlabel('Time'); ylabel('Sample Paths');grid on; hold on;
plot(t' , Y(:,1:100));
set(gcf,'color','white');
fig_no = fig_no + 1 ;

```

A.6 M6. Figure 4.6

```

%% Euler Sumilation for varying step sizes
seed = randn; mu1 = 1.5; sigma1= 2; Z(1,1)= 0 ; T = 1;

% for N=10;
N=10; h = T/N; Z = randn(N,1);Y1 = NaN(N,1); Y1(1,1) = seed;

for i=1:N-1
Y1(i+1,1) = Y1(i,1)+ Y1(i,1)* mu1 *h  + (Z(i+1,1)) * sigma1 * Y1(i,1)*sqrt(h);
end
t1 = linspace(0,N,N);

%% for N=100;
N=100; h = T/N; Z = randn(N,1);Y2 = NaN(N,1); Y2(1,1) = seed;

for i=1:N-1
Y2(i+1,1) = Y2(i,1)+ Y2(i,1)* mu1 *h  + (Z(i+1,1)) * sigma1 * Y2(i,1)*sqrt(h);
end
t2 = linspace(0,N,N);
%% for N=1000;
N=1000; h = T/N; Z = randn(N,1);Y3 = NaN(N,1); Y3(1,1) = seed;

for i=1:N-1
Y3(i+1,1) = Y3(i,1)+ Y3(i,1)* mu1 *h  + (Z(i+1,1)) * sigma1 * Y3(i,1)*sqrt(h);
end
t3 = linspace(0,N,N);
%% for N = 10000;

```

```

N=10000; h = T/N; Z = randn(N,1); Y4 = NaN(N,1); Y4(1,1) = seed;

for i=1:N-1
Y4(i+1,1) = Y4(i,1)+ Y4(i,1)* mu1 *h + (Z(i+1,1)) * sigma1 * Y4(i,1)*sqrt(h);
end
t4 = linspace(0,N,N);

%% Plotting for diff values of h

figure(fig_no)
title('Euler Simulation ');

subplot(4,1,1)
xlabel('Step Size , h = 0.1'); ylabel('Sample Path');grid on; hold on;
plot(t1' , Y1 );
subplot(4,1,2)
xlabel('Step Size , h = 0.01'); ylabel('Sample Path');grid on; hold on;
plot(t2' , Y2 );
subplot(4,1,3)
xlabel('Step Size , h = 0.001'); ylabel('Sample Path');grid on; hold on;
plot(t3' , Y3 );
subplot(4,1,4)
xlabel('Step Size , h = 0.0001'); ylabel('Sample Path');grid on; hold on;
plot(t4' , Y4 );

set(gcf,'color','white');
fig_no = fig_no + 1;

```

A.7 M7. Monte Carlo Simulation versus Euler Approximation

```

%% Single MC and EULER
seed = randn; mu1 = 1.5; sigma1= 2;T = 1;N=1000;h = T/N; Z = randn(N,1);
Z(1,1)= 0;
% Single Monte-carlo Simulation
X = NaN(N,1); X(1,1) = seed;

for i=1:N-1
    X(i+1,1) = X(i,1)+ mu1 * h + (Z(i+1,1)) * sigma1 * sqrt(h) ;

```

```

end

% Single Euler Simulation
Y = NaN(N,1); Y(1,1) = seed;

for i=1:N-1
Y(i+1,1) = Y(i,1)+ Y(i,1)* mu1 *h  + (Z(i+1,1)) * sigma1 * Y(i,1)*sqrt(h);
end
t = linspace(0,N,N);

figure(fig_no)
subplot(1,2,1)
title('Monte-Carlo Simulation ');
xlabel('Time'); ylabel('Sample Path');grid on; hold on;
plot(t' , X );
subplot(1,2,2)
title(' Euler Simulation ');
xlabel('Time'); ylabel('Sample Paths');grid on; hold on;
plot(t' , Y);
set(gcf,'color','white');
fig_no = fig_no + 1;

```

A.8 M8. Simulation using EWMA

```

%%
%%
%We have selected two closing prices series, namely Citigroup and
%Microsoft, from the excel file named 'Data_Daily.xls' and saved it in the
%matrix data. Columns 11 and 18 in "data" contain the adjusted
% closing prices of Citigroup and Microsoft respectively.

data = xlsread('Data_Daily.xls','Data_Yahoo');
[numrows numcols]= size(data);

t_days = 252;%number of trading days per year
data_start=1990;
data_end=2010;
Time_vec=linspace(data_start,data_end,numrows);

%----- We first calculate the log returns

```

```

num_data_pts=numrows;num_stocks=2;
logret= zeros(numrows,2)*NaN;

%calculating the log returns
logret(1,:)=0;
for i=2:num_data_pts
    logret(i,1)=log((data(i,11)/data(i-1,11)));
    logret(i,2)=log((data(i,18)/data(i-1,18)));
end

logret(:,1)=logret(:,1)- mean(logret(:,1)); %demeaning the log returns
logret(:,2)=logret(:,2)- mean(logret(:,2));
Mean = [mean(logret(:,1)) mean(logret(:,2))];

%----- Plotting the log returns

figure(fig_no);
subplot(1,2,1)
title('log returns of citi');
xlabel('Time'); ylabel('Log returns');grid on; hold on;
plot(Time_vec,logret(:,1),'LineStyle','-','LineWidth',1,'Color','blue');

subplot(1,2,2)
grid on; hold on;
title('log returns of microsoft');
xlabel('Time'); ylabel('Log returns');
plot(Time_vec,logret(:,2),'LineStyle','-','LineWidth',1,'Color','red');

set(gcf,'color','white');
fig_no = fig_no + 1;

%% Using EWMA to obtain the time-varying volatilities:
EWMA_ht=NaN(numrows,3);%we take 3 since we have 2 return series,hence
%there'll be 2 variances(citi and then msft) and 1 covariance term
lambda=0.94;
EWMA_cov_mat=cov(logret);
EWMA_ht(1,:)=EWMA_cov_mat([1,4,2]); %the first row of
% the conditional volatility matrix contains the variances/
%volatilities of citi,msft and their covariance resp

```

```

for i=2:numrows
    EWMA_cov_mat=lambda*EWMA_cov_mat +(1-lambda)* logret(i-1,:)'*logret(i-1,:);
    EWMA_ht(i,:)=EWMA_cov_mat([1,4,2]);% EWMA_ht is a 5295x3 matrix, the first col
    %and second columns of EWMA_ht contain the volatilities
    %of citi and msft resp and the third column contains the covariance
end
EWMA_rho=EWMA_ht(:,3)./sqrt(EWMA_ht(:,1).*EWMA_ht(:,2)); %5295x1 matrix
EWMA_res1=logret(:,1)./sqrt(EWMA_ht(:,1));
EWMA_res2=logret(:,2)./sqrt(EWMA_ht(:,2));

%%
mu1 = 1.5; sigma1= 2;
seed = randn;
Mu = mu1 * logret(:,1);
Sigma = sigma1 * logret(:,1);

N=numrows; T = 10; h = T/numrows; Z = randn(N,1); Z(1,1)= 0;
X = zeros(N,1); X(1,1) = seed;

for i=1:N-1
    X(i+1,1) = X(i,1)+ Mu(i,1) * h + (Z(i+1,1)) * Sigma(i,1) * sqrt(h);
end
t = linspace(0,N,N);

figure(fig_no)
title('MC Simulation for the GBM model using  $u(t,S_t) = \mu * S_t$  and
 $v(t,X_t) = \sigma * S_t$ , where  $\mu = 1.5$ ,  $\sigma = 1$  ');

xlabel('Time'); ylabel('Sample Path of GBM model');grid on; hold on;
plot(t' , X );
set(gcf,'color','white');
fig_no = fig_no + 1;

% GBM model (MC Simulation) With Varying volatilities estimated using EWMA
% The same is done by creating a vector of Sigma and Mu by the using the time
% Varying volatilities EWMA_ht in the vector Sigma.

```

Bibliography

- [1] A.F. Ivanov and A.V. Swishchuk (2008), Optimal control of stochastic differential delay equations with application in economics. International Journal of Qualitative Theory of Differential Equations and Applications 2, no. 2 , 201-213.
- [2] Albert N. Shiryaev (2012), Steklov Mathematical Institute, Lectures on some specific topics of Stochastic Calculus and its methods at the London School of Economics
- [3] Andreas Klimke (2002), Scientific/Educational MATLAB Database, Weierstrass
- [4] Bernt Øksendal , Stochastic Differential Equations : An introduction with Applications, 6th Edition, Springer.
- [5] Bernt Øksendal, Agns Sulem , Tusheng Zhang (January 2011), Optimal control of stochastic delay equations and time-advanced backward stochastic differential equations, Stochastic Methods and models, Institut National De Recherche en informatique et en automatique , No 7518
- [6] Bo Martin Bibby, Ib Michael Skovgaard and Michael Sørensen, Diffusion type models with given marginal distribution and autocorrelation function (2005), International Statistical Institute (ISI) and Bernoulli Society for Mathematical Statistics and Probability,191-220, <http://www.jstor.org/stable/3318925>
- [7] David Gamarnik, Advanced Stochastic Processes Lecture Notes, Massachusetts Institute of Technology
- [8] David Williams, Probability with Martingales, Cambridge University Press

- [9] Desmond J. Higham (2001), An Algorithmic Introduction to Numerical Simulation of Stochastic Differential Equations, Society for Industrial and Applied Mathematics Review, Vol. 43, No.3, p . 525-546
- [10] Eckhard Platen and Nicola Bruti-Liberati , Numerical Solution of Stochastic Differential Equations with jumps in Finance, Springer
- [11] F.P.Ramsey (Dec 1928), A Mathematical Theory of Saving, The Economic Journal, Vol.38, No. 152, 543-559
- [12] G.C. Papanicolaou (May, 1973), Stochastic Equations and their Applications, The American Mathematical Monthly, Vol. 80, No. 5 , pp. 526-545
- [13] Hector J. Sussmann (Feb, 1978), On the Gap Between Deterministic and Stochastic Ordinary Differential Equations, The Annals of Probability, Vol. 6, No. 1, pp 19-41
- [14] Jon Danielsson, Financial Risk Forecasting, The theory and practice of forecasting market risk, with implementation in R and MATLAB, Wiley Finance
- [15] Kiyosi Itô and Makiko Nisio, On stationary solutions of a stochastic differential equation, J. Math. Kyoto Univ., Volume 4, Number 1 (1964), 1-75.
- [16] Martin Haugh (2004), Simulating Stochastic Differential Equations, Monte Carlo Simulation: IEOR E4703, Columbia University
- [17] Martin Haugh (2010), Introduction to Stochastic Calculus, IEOR E4707: Financial Engineering: Continuous-Time Models Lecture notes, Columbia University
- [18] Nick Bingham, Stochastic Calculus, Lecture notes for Stochastic Calculus, MA414, Stochastic Analysis, Department of Mathematics, London School of Economics and Political Science
- [19] Paul Glasserman, Monte Carlo methods in Financial Engineering, Springer
- [20] Peter E. Kloeden, Eckhard Platen and Henri Schurz, Numerical Solution of Stochastic Differential Equations through computer experiments.
- [21] Peter E.Kloeden and Eckhard Platen, Numerical Solution of Stochastic Differential Equations , Springer

- [22] Philip E. Protter, Stochastic Integration and Differential Equations, Second Edition, Version 2.1, Springer
- [23] Robert C. Merton (January-February 1976), Option Pricing When Underlying Stock Returns are Discontinuous, Journal of Financial Economics 3, 125-144.
- [24] Robert Jarrow and Philip Protter, A short history of stochastic integration and mathematical finance the early years, 1880-1970, IMS Lecture Notes Monograph, Vol. 45 (2004) 1-17.
- [25] Susanne Ditlevsen, Stochastic differential equation models in biology, Lecture notes for Parameter Estimation for Stochastic Differential Equations, Parameter Estimation in physiological models, Euro Summer School Lipari
- [26] Umut Cetin, Lecture Notes, ST409, Department of Statistics, London School of Economics and Political Science
- [27] Umut Cetin, Robert Jarrow, Philip Protter and Yildiray Yildirim (2004), Modelling credit risk with partial information, The Annals of Applied Probability, Vol. 14, No. 3, 1167-1178